

Quality, Necessity, or Accessibility? Impact of Dockless Bike Sharing on Ineffective Use of Medical Resources

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Abstract

In recent years, patients have increasing autonomy in medical decision-making. With great autonomy in selecting medical institutions, patients with low-risk diseases may opt for treatment in prestigious hospitals, which limits the access of patients with high-risk diseases to these hospitals. Moreover, due to limited medical knowledge and a fear of uncertainty regarding diseases, patients with greater medical autonomy may request various unnecessary medical tests to alleviate their health-related anxiety. This study takes a novel patients' perspective to investigate how dockless bike sharing, which addresses the "last-mile problem" and improves geographical accessibility to medical institutions, affects two types of ineffective use of medical resources: disease-institution mismatch and overuse of medical tests. By introducing the concept of quality-accessibility trade-off in patients' decision-making processes, we shed light on the psychological mechanism underlying the effect of dockless bike sharing on patients' medical institution choices. We also introduce a new concept, perceived necessity of medical tests, to explain how dockless bike sharing reduces unnecessary medical tests. In Study 1, we designed a quasi-experiment and applied a difference-in-differences model based on real-world electronic health record data to show that dockless bike sharing increased the likelihood of patients choosing hospitals over clinics and decreased the number of unnecessary medical tests. In Studies 2a, 2b, 3a, and 3b, four randomized controlled experiments were employed to replicate the findings and reveal the underlying mechanisms. In sum, we found that dockless bike sharing acted as a "double-edged sword" in terms of its impact on exacerbating disease-institution mismatch and alleviating the overuse of medical tests. This study makes significant theoretical contributions to the Operations Management, sharing economy, and medical resource management literature. It also offers important managerial insights for governments, shared mobility platforms, and healthcare providers.

Keywords

Disease-Institution Mismatch, Overuse of Medical Tests, Dockless Bike Sharing, Quality-Accessibility Trade-Off, Perceived Necessity of Medical Tests

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1 Introduction

In recent years, patients have increasing autonomy in medical decision-making, for example, choosing a medical institution and requesting medical tests (Kilbride and Joffe, 2018). Although increased autonomy facilitates shared decision-making and reconceptualizes the patient–physician relationship, this may also bring practical problems, such as ineffective use of medical resources (Brett and McCullough, 2012; Davies and Elwyn, 2008). First, with great autonomy in selecting medical institutions, patients with low-risk diseases (LRD) may opt for treatment in prestigious hospitals, which limits the access of patients with high-risk diseases (HRD) to these hospitals.

The issue of disease-institution mismatch is especially evident in countries without a "gatekeeper" medical referral system,

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where patients have unrestricted freedom to select the medical institutions that they wish to visit (Reibling and Wendt, 2012). Second, due to limited medical knowledge and a fear of uncertainty regarding diseases, patients with greater medical autonomy may request various unnecessary medical tests to alleviate their health-related anxiety, even when experiencing only mild symptoms (Kravitz, 2001). According to a recent survey, 43.2% of Chinese medical staff acknowledge the overuse of medical tests, with a significant portion of these being requested by patients (Sun and Shen, 2020).

Given the increasingly important role of patients in medical decision-making, it is critical to understand the psychological mechanisms behind their choices of medical institutions and requests for additional medical tests. The literature has shown that choosing a medical institution involves balancing the quality of medical services with the accessibility to hospitals/clinics (Shukla et al., 2021). In prior studies, accessibility is typically defined by factors such as appointment availability, convenient office hours, and waiting times (e.g., Keskinocak and Savva, 2020; Patrick, 2011). However, geographical accessibility—referring to the time and effort required to travel to medical institutions—has received less attention in the literature on medical decision-making. Moreover, when patients request additional medical tests, quality is no longer a major concern due to standardized procedures, but geographical accessibility continues to play a crucial role in shaping their decision-making. Therefore, this research explores the underlying mechanisms by which improved geographical accessibility affects two types of ineffective medical resource use among patients with LRD: disease-institution mismatch and overuse of medical tests.

In urban areas, patients with LRD often rely on public transportation to reach medical institutions, given their relatively mild symptoms and the challenges of hospital parking. Like other public transportation users, they frequently encounter the “last-mile problem,” which arises when public transit only provides station-to-station commuting and leaves commuting from a transit station to the destination disproportionately costly in terms of time and money (Chu et al., 2021). Recent advancements in information technologies (e.g., mobile payments, global positioning systems, and smart locks) have made dockless bike sharing an effective solution to the “last-mile problem” of public transportation (Jin et al., 2024). As an innovative and cost-effective mode of urban commuting, dockless bike sharing offers several advantages over traditional transportation methods, including flexible mobility, reduced travel time, one-way trip possibility, and multimodal transport connection (He et al., 2021; Kabra et al., 2020). In Chinese cities, dockless bike sharing has become a popular and convenient transportation option for patients with LRD to reach medical institutions. According to the China Road Transport Association, more than 47 million daily trips were made using dockless bike-sharing services in 2021,¹ a considerable portion of which was for the purpose of accessing medical institutions. Reports indicate that the extensive use of dockless bike sharing

has led to a significant accumulation of bikes parked in a way that blocks the entrances of prestigious hospitals² (see Figure G1 in E-companion G).

Although the Operations Management (OM) literature has indicated that dockless bike sharing can improve geographical accessibility and impact diverse outcomes (e.g., Chu et al., 2021; Jin et al., 2024), there is limited understanding of its influence on patients with LRD’s ineffective use of medical resources. First, the decision-making process of choosing a medical institution involves a trade-off between the quality of medical services and the geographical accessibility to medical institutions (Shukla et al., 2021). Patients generally prefer hospitals (i.e., higher-ranked hospitals) over clinics (i.e., lower-ranked hospitals) when the quality of medical services is prioritized. However, when geographical accessibility is a key consideration, patients with LRD may opt for a nearby clinic instead of a distant hospital. It remains unknown how dockless bike sharing, which addresses the “last-mile problem” in accessing medical institutions, would reshape the quality-accessibility trade-off and influence the choices of medical institutions for patients with LRD. Thus, our first research question is: *How does dockless bike sharing affect patients with LRD’s choices of medical institutions?*

Moreover, patients request additional medical tests based on their perceived necessity, which encompasses their assessment of the appropriateness and cost-effectiveness of performing a particular medical test (Glassman et al., 1997; Kravitz, 2001). This perceived necessity reflects patients’ subjective judgment and may not align with the objective necessity determined by medical standards. Notably, changes in geographical accessibility can influence patients’ perceived necessity of medical tests. Specifically, when accessing a medical institution is inconvenient, patients may perceive a medical test as more necessary and are likely to request more medical tests during their current visit to avoid potential future trips. In contrast, when patients have high geographical accessibility to a medical institution, the same medical test may seem less urgent, leading them to undergo fewer medical tests during the current visit and postpone less necessary ones for future visits. Despite the role of dockless bike sharing in addressing the “last-mile problem,” none of the literature has investigated its influence on the overuse of medical tests through geographical accessibility and perceived necessity of medical tests. Thus, our second research question is: *How does dockless bike sharing affect the number of medical tests conducted by patients with LRD in a single visit?*

We leveraged a mixed-method design to answer the research questions and unravel the underlying mechanisms behind the effects. Specifically, Study 1 constituted a quasi-experiment to examine the impact of dockless bike sharing on the ineffective use of medical resources based on a real-world dataset of patients’ electronic health records (EHR). Taking advantage of the entry of dockless bike sharing into a district in Zhejiang province of China in September 2017, we used a difference-in-differences (DID) method to examine the causal

effects of dockless bike sharing on patients' choice of medical institutions and the number of medical tests conducted in a single visit. In Studies 2a and 2b, two randomized controlled experiments were employed to validate the findings regarding the effect of dockless bike sharing on the choice of medical institutions and its underlying mechanism. In Studies 3a and 3b, two additional randomized experiments were conducted to test the relationship between dockless bike sharing and the number of unnecessary medical tests, as well as to examine the underlying mechanism.

2 Literature Review

2.1 Disease-Institution Mismatch

The OM literature shows that mismatches frequently occur in medical systems due to the capacity constraints of medical resources (Chan et al., 2019; Song et al., 2020). Researchers have focused on two types of mismatches in medical systems: off-level placement and off-service placement. The former relates to the mismatch between the severity of a patient's disease and the level of care intensity that the patient received (e.g., a patient who needs a high level of care intensity is placed in a unit with a lower level of care intensity) (Bretthauer et al., 2011; Liu et al., 2019). The latter refers to the situation in which a patient is placed in an open bed in a unit designated for a different service because the beds in the appropriate unit are at full capacity (Song et al., 2020). To the best of our knowledge, little attention has been paid to the mismatch between the severity of a patient's disease and the medical institution that the patient visits.

The issue of disease-institution mismatch is particularly pronounced in countries lacking a medical referral system. The medical referral system enforces the separation of primary and secondary care by introducing a "gatekeeper," such as a general practitioner, to ensure that patients seek specialized care only when necessary (Sweeney, 1994). In countries without a medical referral system, patients can bypass the usual "gatekeepers" in primary care and have great autonomy in choosing medical institutions. This means they can choose to visit either a higher-ranked hospital providing comprehensive medical services or a lower-ranked hospital offering fundamental medical treatment. Disease-institution mismatch may arise when patients with LRD prioritize the quality of medical services and patients with HRD face reduced accessibility to higher-ranked hospitals (Li et al., 2021). Such mismatch may lead to low cost-effectiveness for patients with LRD and increased mortality risk for patients with HRD (Liu et al., 2017). For convenience, we use the term "hospitals" to represent higher-ranked hospitals (i.e., level 2 and level 3 hospitals) and the term "clinics" to represent lower-ranked hospitals (i.e., level 1 hospitals) in this paper.

2.2 Overuse of Medical Tests

Overuse of medical tests refers to the unwarranted/unnecessary conduct of medical tests that are unlikely

to provide additional clinical benefits or carry risks of harm exceeding benefits (Brett and McCullough, 2012; MacLeod et al., 2018). Performing unnecessary medical tests can lead to a variety of negative consequences, such as unnecessary sample collection, prolonged hospital stays, adverse health outcomes, and the waste of healthcare resources (Kim et al., 2011; Young et al., 2023). Several approaches have been proposed to mitigate the overuse of medical tests from the perspective of medical service providers, including redesigning pathology request forms, imposing limits on test orders, requiring pathologist preapproval, and enabling healthcare information sharing (Adjerid et al., 2018; Ayabakan et al., 2017; Bailey et al., 2005; May et al., 2006). However, May et al. (2006) indicate that changes made on the physicians' side only have a temporary impact on reducing the overuse of medical tests, as they lack continuous motivation or incentive rewards to sustain such changes.

With the increasing autonomy of patients in medical decision-making, a substantial number of medical tests are now conducted at their request (Quill and Brody, 1996). In many cases, physicians serve as consultants to offer a range of options based on patients' health conditions and preferences, while patients decide which medical tests they wish to undergo (Kilbride and Joffe, 2018). Due to their limited professional medical knowledge, patients are likely to request medical tests that are not clinically necessary. Under such circumstances, some physicians may choose not to decline these requests, either to satisfy patients' demands or to enhance the hospital's revenue (Brett and McCullough, 2012). Considering the growing trend of unnecessary medical tests requested by patients, our research takes a novel patient-centered perspective to explore the overuse of medical tests.

2.3 Dockless Bike Sharing

As dockless shared bikes can be operated without docking stations and parked anywhere, they can facilitate neighborhood accessibility by functioning as a "last-mile" connection and effectively solve the "last-mile problem" (Kabra et al., 2020). Specifically, dockless shared bikes can serve as a supplement to public transit, and be used for commuting between a bus or subway station and one's intended destination (Jin et al., 2024). The integration of dockless bike sharing with public transportation fundamentally changes the commuting practices of individuals and constitutes a novel mode of travel. Furthermore, compared to conventional modes such as buses and cars, the relatively low transportation cost of dockless bike sharing also contributes to its popularity in countries like China, where individuals can gain unlimited access to dockless bike-sharing services after paying a monthly fee of less than 2 US dollars. Due to the convenience and cost-effectiveness of dockless bike sharing, a significant proportion of trips previously made by car or on foot are now replaced by dockless bike-sharing trips (Jin et al., 2024; Qin et al., 2018).

The OM literature has primarily focused on dockless shared bikes' supply management strategies, such as bike

repositioning, regulatory policies (e.g., geo-fencing), and user incentives (e.g., crowdfunding) (Freund et al., 2018; He et al., 2021; Qi and Shen, 2019). Other studies have explored the externality of dockless bike sharing. For instance, it expands employment options by providing daily commuters a more convenient means of transportation (Bullock et al., 2017). Moreover, Chu et al. (2021) found that the introduction of dockless bike sharing decreases the housing price premium by 29% per kilometer away from a subway station. In addition, Jin et al. (2024) observed that dockless bike sharing, unlike ride sharing or traditional bike sharing, augments subway usage, serving as a complement to public transportation rather than a substitute.

Notably, dockless shared bikes have been commonly used by patients with LRD as a convenient and cost-effective mode of transportation to access medical institutions. Although a large number of studies have shown that dockless bike sharing can improve geographical accessibility and address the “last-mile problem,” none of these studies have shed light on the impact of dockless bike sharing on the use of medical resources. Unlike outcomes such as housing prices or public transportation usage, patients’ medical decision-making is shaped by more complex psychological dynamics, including quality-accessibility trade-offs and perceived necessity of medical tests—an area that has received less attention in the OM literature. Given the widespread concern over the ineffective use of medical resources, this research uncovers the psychological mechanisms driving the effect of dockless bike sharing on patients’ medical choices.

3 Hypothesis Development

3.1 Dockless Bike Sharing and Medical Institution Choice

To understand how dockless bike sharing influences disease-institution mismatch, we explore the psychological mechanism of quality-accessibility trade-off in patients’ choices of medical institutions. Previous research has identified quality of medical care and accessibility to medical services as the two most significant factors affecting patients’ choices of medical services (Razzouk et al., 2004). On the one hand, patients need to consider the quality of medical services as it greatly impacts their health outcomes (Chan et al., 2019). For example, Shukla et al. (2021) show that patients are willing to see doctors who provide high-quality medical services according to digital word-of-mouth even if they need to travel farther. On the other hand, patients also need to consider accessibility because high accessibility enables timely treatment and convenient visits. Prior OM studies have shown that accessibility to medical institutions is a key consideration in healthcare operations (de Vries et al., 2020; Tippong et al., 2022). When accessibility is low, patients may experience delays in receiving medical care, resulting in detrimental health outcomes (Lu and Lu, 2018).

Since few medical institutions can satisfy both criteria simultaneously, patients often have to make a trade-off between quality and accessibility (Ferreira and Marques, 2019). Unlike other aspects of accessibility, such as appointment availability, convenient office hours, and waiting times (e.g., Keskinocak and Savva, 2020; Patrick, 2011), geographical accessibility to medical institutions has received less attention in the literature on patients’ medical decision-making. Given that the significant time and effort required for travel to medical institutions can discourage patients with LRD from seeking high-quality medical treatment, our research examines how enhanced geographical accessibility, facilitated by dockless bike sharing, affects patients’ quality-accessibility trade-off, which ultimately influences their choice of hospitals/clinics.

In China, due to the scarcity of high-quality medical resources, there are fewer hospitals than clinics. While hospitals are usually located in the center of a district, clinics are widely distributed throughout local communities. Therefore, patients are generally farther from hospitals than clinics. Before the entry of dockless bike sharing, patients often face the “last-mile problem” when traveling to hospitals, which typically requires a substantial amount of time and effort. By contrast, it is much more convenient for them to find a clinic in their communities. Under such circumstances, low geographical accessibility to hospitals may divert some patients with LRD to choose clinics near their community.

The entry of dockless bike sharing, however, reduces the travel costs of visiting medical institutions, especially enhancing the geographical accessibility to hospitals by addressing the “last-mile problem.” Compared to hospitals, the improvement in geographical accessibility to clinics is limited because patients usually do not have the “last-mile problem” to visit clinics. Given the reduced gap in geographical accessibility to hospitals and clinics after the introduction of dockless bike sharing, patients with LRD tend to put less weight on accessibility and attach more importance to quality in their medical institution choice. For this reason, patients with LRD are more likely to choose hospitals instead of clinics after the entry of dockless bike sharing.³ Therefore, we hypothesize the following:

H1: The entry of dockless bike sharing increases the likelihood that patients with LRD choose to go to hospitals rather than clinics.

3.2 Dockless Bike Sharing and Overuse of Medical Tests

Given the adverse consequences of performing unnecessary medical tests (Kim et al., 2011; Young et al., 2023), it is preferable for patients to receive medical tests at the appropriate time, rather than completing all medical tests in a single hospital visit. This study introduces a novel concept, perceived necessity of medical tests, to understand how dockless bike sharing influences the overuse of medical tests. It is

worth noting that perceived necessity is a subjective construct that may be influenced by other factors, such as geographical accessibility to a medical institution. If the geographical accessibility to a medical institution is low, traveling there would be time-consuming and exhausting. Under such circumstances, patients may perceive unnecessary medical tests as more essential. To avoid making multiple trips, they may opt to undergo unnecessary medical tests during their current visit, even though these tests could be postponed to future appointments. In contrast, if the geographical accessibility to a medical institution is high, patients may be less concerned about the need for repeat visits. As a result, they may perceive the necessity of the same medical test to be lower. Accordingly, they are more likely to undergo only the necessary medical tests during each visit and defer any unnecessary tests to future visits.

Dockless bike sharing improves accessibility to medical institutions, which further influences patients' decisions on which medical tests to take during the current visit. Specifically, before the entry of dockless bike sharing, because geographical accessibility to medical institutions is low, patients may take some unnecessary medical tests in the current visit to avoid additional visits. The advent of dockless bike sharing, however, promotes patients' geographical accessibility to medical institutions. Because high geographical accessibility gives patients the impression that they can easily revisit the medical institution, they are more likely to view optional medical tests as less necessary. Under such circumstances, patients may choose to take only essential medical tests in each visit, deferring superfluous tests to future visits. As treatment progresses, it is very likely that patients have recovered from the illness or gained a better understanding of their health conditions. As a result, they may not visit the medical institution again. Even if they do return, they would realize the deferred medical tests are unnecessary and opt not to undergo them. Thus, deferring superfluous tests to future visits can result in a reduction of unnecessary tests. By avoiding unnecessary medical testing, the number of medical tests received by patients with LRD during a single visit decreases after the entry of dockless bike sharing. Therefore, we hypothesize the following:

H2: The entry of dockless bike sharing decreases the number of medical tests that patients with LRD receive in a single visit.

4 Study I

4.1 Research Context and Data

To examine the impact of dockless bike sharing on the choice of medical institutions and the number of medical tests, we set our research context in a district in Zhejiang province of China, where dockless bike-sharing services started in September 2017. There were 26 clinics and three hospitals in the focal district. All of the clinics were located near residential communities to provide convenient medical services. Due to the

limited number of hospitals in this district, it took much more time and effort for patients to go to hospitals. However, as hospitals could provide higher-quality medical services than clinics, some patients still preferred to go to hospitals even if they had LRD.

We obtained patients' EHR in this district by collaborating with the local government. The dataset contained each patient's demographics (e.g., age and gender), medical registration and consultation information (e.g., medical institution visited, symptoms, and doctor diagnosis), and medical test information (e.g., medical test type, medical test item, and result). We aggregated the data at the individual-visit level. Our analysis sample included 177,113 visits made by 72,704 patients with LRD, spanning from 1 March 2017 to 30 September 2018.

We hired three experts from a major medical school in China to manually label the severity of diseases in our dataset. All diseases were classified into three categories: HRD (i.e., diseases listed in China Severe Disease Insurance,⁴ for example, cancer, organ transplantation, and cardiomyopathy), LRD (e.g., bronchitis, gastritis, and rhinitis), and other diseases (i.e., diseases that were too general to assess the severity, for example, diabetes and hepatitis B). The severity level of a disease was confirmed if at least two experts agreed on the label; otherwise, experts would have a discussion to reach a consensus on the severity of the disease. The detailed types of LRD are listed in E-Companion A1.

4.2 Quasi-Experiment Design and DID Estimation

We designed a quasi-experiment and employed a DID estimation to investigate the effects of dockless bike sharing on the ineffective use of medical resources. Exogenous governmental approval of dockless bike sharing in the focal district in September 2017 provided an ideal quasi-experimental setting. Importantly, the launch of dockless bike-sharing services was unlikely to be driven by medical resource usage. However, the availability of dockless bike-sharing services was likely to improve geographical accessibility to medical institutions.

In the quasi-experiment, we selected patients with LRD aged from 18 to 25 as the treatment group and patients with LRD aged from 5 to 12 as the control group based on the following reasons. First, individuals aged from 18 to 25 were open to new technologies and faced greater financial constraints than other age groups. Thus, this group was highly inclined to use dockless bike-sharing services to go to medical institutions if they had LRD. Second, in accordance with Chinese legislation, individuals below the age of 12 were prohibited from riding dockless shared bikes. As a result, their medical resource usage was not influenced by the availability of dockless bike sharing. Because shared bikes can accommodate only one rider at a time, it is not feasible for parents to utilize a shared bike to accompany their children for medical visits. Third, reports have shown that both the treatment and control groups⁵ exhibit comparable morbidity rates.⁶ There is

no evidence suggesting any disparities in these two groups' preferences for medical institutions or inclination to request unnecessary medical tests. In addition, we conducted a relative time model with leads and lags time dummies to show that the treatment and control groups had parallel trends in the dependent variables before the entry of dockless bike sharing (see the Alternative Identification Strategy: Relative Time Model section for details).

To estimate the effects of dockless bike sharing on the ineffective use of medical resources, we used the DID approach to compare the changes in patients' use of medical resources after the availability of dockless bike sharing between the treatment and control groups. DID was suitable in our research context because medical resource usage of the treatment group (i.e., patients with LRD aged from 18 to 25) was affected by the entry of dockless bike sharing, while that of the control group (i.e., patients with LRD aged from 5 to 12) was not. DID estimation implicitly controlled for any bias caused by variables common to the treatment and control groups (Meyer, 1995) and allowed us to examine a causal relationship between dockless bike sharing and the ineffective use of medical resources. Moreover, potential confounding factors, such as parental involvement in medical decisions for patients under 12, were eliminated in the DID estimation by examining the changes in medical resource usage after the entry of dockless bike sharing. Considering that it took some time for dockless bike sharing to gain popularity (Chu et al., 2021), we followed Furman and Stern (2011)'s approach to allow for an initial one-month burn-in period (September 2017). Therefore, March 2017 to August 2017 was the pre-treatment period, and October 2017 to September 2018 was the post-treatment period.⁷

4.3 Econometric Model Specifications

To test the effect of dockless bike sharing on patients' choice of medical institutions (H1), we set up the DID model as follows:

$$HospVisit_{i,t} = \alpha_i + \delta_t + \beta_1 \cdot Treatment_i \times EntryMonth_t + \beta_2 \cdot Treatment_i \times PostEntry_t + \epsilon_{i,t} \quad (1)$$

where the dependent variable $HospVisit_{i,t}$ was a binary variable indicating whether the medical institution that patient i visited at time t was a hospital ("1" for hospital; "0" for clinic); α_i captured the individual fixed effects controlling for any time-invariant omitted factors that characterized each individual; δ_t was the month-of-year fixed effects, the inclusion of which allowed us to control potential time-specific idiosyncrasies in this district, such as seasonal fluctuation of diseases and temperature; $Treatment_i$ was a binary variable with a value of "1" if patient i belonged to the treatment group (patients aged from 18 to 25), and "0" if patient i belonged to the control group (patients aged from 5 to 12); $EntryMonth_t$ was a binary indicator with a value of "1" if time t was in September 2017 and "0" otherwise; $PostEntry_t$ was a binary indicator with a

value of "1" if time t was in the post-treatment period (from October 2017 to September 2018) and "0" otherwise. In the model specification, $Treatment_i$ was omitted because of individual fixed effects, while $EntryMonth_t$ and $PostEntry_t$ were absorbed because of month-of-year fixed effects. Our primary interest was β_2 which captured the estimated effect of dockless bike sharing on patients' choice of medical institutions. Since our dependent variable ($HospVisit_{i,t}$) was a binary variable, we employed a fixed-effects logit estimator.

To test the effects of dockless bike sharing on the number of medical tests (H2), we set up the DID model as follows:

$$NumofTests_{i,t} = \alpha_i + \delta_t + \beta_1 \cdot Treatment_i \times EntryMonth_t + \beta_2 \cdot Treatment_i \times PostEntry_t + \epsilon_{i,t} \quad (2)$$

where the dependent variable $NumofTests_{i,t}$ was the number of medical tests that patient i received in the visit at time t . The other variables remained the same as those in Econometric Model Specification (1). Of our primary interest was β_2 which captured the estimated effect of dockless bike sharing on the number of medical tests conducted in a single visit. Since the dependent variable $NumofTests_{i,t}$ was in the form of counts, we employed a fixed-effects Poisson pseudo-maximum likelihood (PPML) estimator, which has been shown to provide consistent robust standard errors (Burtch et al., 2018; Wooldridge, 1999). Table 1 summarizes the descriptive statistics of all of our variables.

4.4 Results and Discussion

Table 2 presents the results of the relationship between dockless bike sharing and the ineffective use of medical resources. Model (1) shows that dockless bike sharing had a significant effect on patients with LRD's choice of medical institutions ($\beta_2 = .297, p < .001$). Specifically, after the availability of dockless bike sharing, the odds of patients with LRD choosing hospitals were 1.346 times that without dockless bike sharing (see E-Companion A2 for a back-of-envelope calculation and E-companion A3 for an additional analysis regarding the impact of dockless bike sharing on the number of hospital and clinic visits). Therefore, H1 was supported. While previous studies have shown the positive influences of dockless bike sharing on complementing public transportation (Jin et al., 2024) and reducing housing price premium (Chu et al., 2021), our findings contribute to the literature by demonstrating that dockless bike sharing may also aggravate the disease-institution mismatch through improved geographical accessibility. Thus, it is important for bike-sharing platforms to take certain measures to avoid such unintended consequences. Following He et al.'s (2021) approach, a feasible solution could involve designating no-parking zones and charging additional parking fees for violations, such as parking near hospital entrances.

Model (2) reports the overall effect of dockless bike sharing on the number of medical tests in both hospitals and clinics.

Table 1. Descriptive statistics.

Variable	Observations	Mean	SD	Min	Max	1	2	3	4	5	6	7
1. HospVisit	177,113	0.61	0.49	0	1	1						
2. NumofTests	177,113	1.51	0.93	0	7	0.17	1					
3. EntryMonth	177,113	0.05	0.21	0	1	−0.00	0.01	1				
4. PostEntry	177,113	0.67	0.47	0	1	0.00	−0.01	−0.32	1			
5. Treatment	177,113	0.51	0.50	0	1	−0.05	0.19	0.03	−0.02	1		
6. Age	177,113	15.06	7.70	5	25	−0.06	0.19	0.03	−0.02	0.96	1	
7. Gender	177,113	0.45	0.50	0	1	0.04	−0.01	−0.00	0.00	−0.18	−0.18	1

Table 2. Main results.

Variable	(1) HospVisit	(2) NumofTests	(3) NumofTests (Hospitals)	(4) NumofTests (Clinics)
Treatment × EntryMonth	0.019 (0.120)	0.001 (0.020)	−0.010 (0.027)	−0.013 (0.029)
Treatment × PostEntry	0.297** (0.058)	−0.015 (0.010)	−0.030* (0.014)	−0.011 (0.015)
Constant		0.438** (0.009)	0.501** (0.012)	0.320** (0.014)
Patient FE	✓	✓	✓	✓
Month-of-year FE	✓	✓	✓	✓
Number of observations	40,475	141,555	87,478	45,934
Number of patients	8425	37,146	22,883	14,545
Estimation model	FE Logit	FE PPML	FE PPML	FE PPML

Note. Standard errors are reported in parentheses, clustered at the patient level to account for potential serial autocorrelation. ** $p < .01$, * $p < .05$. In the estimation of Model (1), 64279 patients (136638 observations) were dropped because of all positive or all negative outcomes. In the estimations of Models (2)–(4), 35558, 21146, and 22555 observations that were either singletons or separated by a fixed effect were dropped, respectively. Model (1) and Model (2) used the entire sample, but the number of observations differed because different estimators were used. Model (3) included only hospital visits, while Model (4) included only clinic visits.

Different from our expectation, the coefficient of the interested interaction term was not significant in Model (2) ($\beta_2 = -.015$, $p = .126$). Considering the mechanism underlying the relationship, the non-significant result might be attributed to the fact that dockless bike sharing improved patients' accessibility to hospitals and clinics to different extents. In our context, clinics were located around residential areas, and thus it was convenient for patients to visit clinics even without dockless bike sharing. As a result, the improvement in geographical accessibility to clinics caused by dockless bike sharing was limited. Thus, when we conducted the analysis based on a sample composed of both hospital and clinic visits, the overall effect of dockless bike sharing on the number of medical tests became non-significant.

To understand the heterogeneity in the effects of dockless bike sharing across hospitals and clinics, we conducted further analyses based on the hospital and clinic subsamples, respectively. Model (3) shows that the availability of dockless bike sharing reduced the average number of medical tests conducted in each hospital visit by 2.95% ($\beta_2 = -.030$, $p = .026$). However, Model (4) shows that the availability of dockless bike sharing had no effect on the average number of medical tests conducted in each clinic visit ($\beta_2 = -.011$, $p = .468$).

Taken together, H2 was partially supported. While previous studies have focused on reducing the overuse of medical tests from the healthcare providers' perspective (e.g., Ayabakan et al., 2017; Young et al., 2023), our research offers a novel approach to addressing the issue by improving patients' geographical accessibility to medical institutions. Following this logic, other forms of shared mobility, such as ride sharing (Diao et al., 2023), can also help mitigate the overuse of medical tests. Moreover, the non-significant effect of dockless bike sharing in the context of clinics suggests that when the last-mile problem is not a major concern, bike sharing does not offer a significant advantage over other transportation options.

4.5 Alternative Identification Strategy: Relative Time Model

We employed a relative time model with leads and lags time dummies as an alternative identification strategy (Burtch et al., 2018). This model not only estimated the impact of dockless bike sharing over pre-entry and post-entry time periods, but also tested the parallel trend assumption for the DID analysis. In the relative time model, we constructed a set of time dummies representing the chronological distance of an observation

period concerning the entering month of dockless bike sharing ($PreEntry_t$, $PostEntry_t$). The relative time model specification is as follows:

$$Y_{i,t} = \alpha_i + \delta_t + \sum_j \sigma_j (Treatment_i \times PreEntry_t) + \varsigma \cdot Treatment_i \times EntryMonth_t + \sum_k \tau_k (Treatment_i \times PostEntry_t) + \varepsilon_{i,t} \quad (3)$$

where $Y_{i,t}$ referred to $HospVisit_{i,t}$ and $NumofTests_{i,t}$, respectively. We focused on the number of medical tests in hospitals only, given that dockless bike sharing had not changed the number of medical tests in clinics. $PreEntry_t$ was the vector of relative time dummies of leads periods, containing indicator variables for months 1–6 before the entry of dockless bike sharing. One month before the entering month was randomly omitted in the analysis, which also served as the baseline. $PostEntry_t$ was the vector of relative time dummies of lags periods, including variables indicating months 1–12 after the entering month of dockless bike sharing. Of our primary interests were σ_j ($j = 1, 2, 3, 5, 6$) and τ_k ($k = 1, 2, 3, \dots, 11, 12$).

Table A2 (see in E-companion A4) shows the results of the relative time models. Model (A3) presents the results of the effect of dockless bike sharing on patients' choice of medical institutions. Model (A4) shows the results of the effect of dockless bike sharing on the number of medical tests in hospitals. In both Models (A3) and (A4), the coefficients of the effects of dockless bike sharing in the pre-entry period were non-significant, indicating that there was no significant difference between the treatment and control groups before the entry of dockless bike sharing. Therefore, the relative time model did not show evidence against the parallel trend assumption for the DID estimation.

For the effect of dockless bike sharing on patients' choice of medical institutions, the coefficients of the interaction term of $Treatment_i$ and post-entry dummies in Model (A3) were significant and positive since the third month after the entry of dockless bike sharing. This was because it took some time for patients to know and accept dockless bike sharing as a more convenient means of transportation. Since most patients preferred to receive high-quality medical services in hospitals, as long as they were aware of the advantage of dockless bike sharing in improving geographical accessibility to hospitals, they would probably choose hospitals. By contrast, regarding the effect of dockless bike sharing on the number of medical tests in hospitals, we found that the coefficients of the interaction term of $Treatment_i$ and post-entry dummies in Model (A4) were significant and negative since the eighth month after the entry of dockless bike sharing.

The lagged effect of dockless bike sharing on the number of medical tests in hospitals might be ascribed to patients' inertia in requesting unnecessary medical tests. That is, even after the introduction of dockless bike sharing, patients may continue

to request unnecessary medical tests out of habit or reluctance to change their usual healthcare-seeking behavior. Over time, as they gradually adapt to the new mode of transportation and the increased geographical accessibility to hospitals, they may reduce the number of unnecessary medical tests received. However, initially, the persistence of these requests can cause a lag in the effect of dockless bike sharing on the number of medical tests. This explains why the effect of dockless bike sharing on the number of medical tests became significant 8 months after the introduction of dockless bike sharing. In sum, we add to the literature on patient behavior (e.g., Dai and Tayur, 2022; Kc and Kim, 2022) by showing that if the changes caused by dockless bike sharing are consistent with patients' preferences (e.g., patients' inclination towards high-quality medical services offered in hospitals), the effect is likely to occur shortly after the entry of dockless bike sharing. In contrast, if the changes caused by dockless bike sharing are inconsistent with patients' preferences (e.g., some patients believe more medical tests can eliminate the risk of misdiagnosis), it usually takes more time for the effect to emerge.

In addition, we performed a series of supplementary analyses to verify our findings. The supplementary analyses are summarized in Table 3 (see E-Companion B, C, and D for details).

5 Study 2a

Although Study 1 demonstrated the effects of dockless bike sharing on patients' ineffective use of medical resources, it did not examine the underlying mechanisms behind the effects because psychological constructs, such as perceived geographical accessibility to medical institutions, quality-accessibility trade-off, and perceived necessity of medical tests, could not be captured based on patients' EHR. The first objective of Study 2a was to replicate the findings of Study 1 under a controlled experimental setting. Study 2a manipulated the availability of dockless bike sharing, which enabled us to infer a causal relationship between dockless bike sharing and patients with LRD's choice of medical institutions. The second goal was to shed light on the underlying mechanism behind this causal relationship. According to our theoretical framework, dockless bike sharing was expected to increase patients with LRD's perceived geographical accessibility to hospitals, which nudged them to prioritize quality in their quality-accessibility trade-off, thereby increasing the chances of visiting hospitals.

5.1 Experiment Design and Procedure

Two hundred and fifty subjects (62% female, $M_{age} = 32.54$) recruited from Credamo (a Chinese crowd-sourcing marketplace that is similar to Amazon Mechanical Turk) participated in Study 2a in exchange for monetary compensation. The participants were randomly assigned to one of two conditions (dockless bike sharing vs. non-dockless bike sharing, *Bike_sharing*) in a single-factor between-subjects design.

Table 3. Summary of supplementary analyses.

No.	Category	Concern	Test	Outcome	Statistical results
1	Alternative explanation elimination	Dockless bike sharing may make patients healthier and reduce the number of medical tests (E-companion B1).	Examining the effect of dockless bike sharing on patients' number of visits to medical institutions per month	The entry of dockless bike sharing did not significantly influence the number of visits to medical institutions, ruling out the "healthier patient" explanation.	Table B1
2		Dockless bike sharing may increase traffic accidents, which in turn augment hospital visits (E-companion B2).		The entry of dockless bike sharing did not significantly influence the number of visits to medical institutions, ruling out the "more traffic accidents" explanation.	
3		More hospital visits made by patients with milder diseases and symptoms lead to fewer medical tests (E-companion B3).	Conducting additional analyses based on the patients who visited hospitals with the same disease and symptoms	Dockless bike sharing still had a negative effect on the number of medical tests in hospitals, ruling out the "milder disease and symptom" explanation.	Table B1
4		The reduced part of medical tests is necessary, instead of unnecessary (E-companion B4).	Restricting the analysis sample to bronchitis patients and manually labeling whether each medical test is necessary Examining the effect of dockless bike sharing on the percentage of unnecessary medical tests in both hospitals and clinics	Consistent with our proposed mechanism, dockless bike sharing led to a reduction in the percentage of unnecessary medical tests in hospitals, but no significant effect was found in clinics.	Table B1
5		Unnecessary medical tests may be simply deferred to future visits (E-companion B5).	Examining whether the total number of medical tests a patient received in six months decreased after the introduction of dockless bike sharing	Dockless bike sharing resulted in a decline in the total number of medical tests in hospitals, while no significant changes in the total number of medical tests were observed in clinics.	Table B2
6	Alternative estimators	The results may be subject to Logit and PPML estimators (E-companion C1).	Employing alternative econometric models, including fixed-effects OLS, random-effects Logit, and random-effects PPML	The results were consistent with the main findings.	Table C1

(continued)

Table 3. Continued.

No.	Category	Concern	Test	Outcome	Statistical results
7	Alternative sample	Different use patterns of medical resources during the Spring Festival (E-companion C2).	Re-estimating models based on the sample, excluding the time period of the Spring Festival of 2018	The results were consistent with the main findings.	Table C1
8	Sensitivity tests	Results may be subject to the setting of the quasi-experiment (E-companion C3).	Re-estimating models based on the alternatively defined treatment (patients aged from 18 to 30, 18 to 35, 18 to 40, and 18 to 45) and control groups (patients aged from 0 to 12 and 60 to 70)	The results were consistent with the main findings. Our findings could be generalizable to different age groups.	Table C2
9			Re-estimating models (1) by defining two-month burn-in periods (August and September 2017/September and October 2017); (2) by incorporating the burn-in period into the post-entry period (September 2017 to September 2018)	The results were consistent with the main findings.	Table C3
10	Falsification tests	Estimated effects of dockless bike sharing may be the consequences of some systematic changes (E-companion D1).	Conducting falsification tests based on patients with HRD	Dockless bike sharing did not have significant effects on the choice of medical institutions or the number of medical tests per hospital visit for patients with HRD.	Table D1
11		Estimated effects of dockless bike sharing may be due to coincidence (E-companion D2).	Conducting nonparametric resampling tests	Our estimated effects of dockless bike sharing stood out in the histograms of what was typically observed among the randomly selected treatment groups.	Figure D1

In both conditions, participants first answered questions about their demographics (i.e., gender, age, education level, and income level). Participants were then asked to imagine a scenario in which they had several symptoms⁸ (i.e., a stuffy nose, runny nose, and cough) and decided to visit a doctor. They were required to recall a hospital and a clinic that they usually visit, and make a choice in this scenario between the hospital and the clinic (*Hosp_choice*). In the *dockless bike-sharing condition*, participants were told that they could use any transportation method (including bus, subway, taxi, bike, dockless shared bike, motorbike, and private car) or walk to go to the hospital or the clinic. In the *non-dockless bike-sharing condition*, participants were told that they could use any transportation methods (including bus, subway, taxi, bike, motorbike, and private car) or walk to go to the hospital or the clinic. The only difference between the two conditions was whether participants were allowed to use dockless bike-sharing services.

Afterward, participants reported their weight of quality in quality-accessibility trade-off (*Trade-off*), perceived geographical accessibility to the hospital (*Hosp_access*), and perceived geographical accessibility to the clinic (*Cli_access*). Participants also rated perceived quality of the hospital (*Hosp_quality*), perceived quality of the clinic (*Cli_quality*), medical literacy (*Med_literacy*), perceived severity of the symptoms in the given scenario (*Severity*), travel time to the hospital (*Hosp_trv_time*), travel time to the clinic (*Cli_trv_time*), and frequency of dockless bike sharing usage (*Bike_freq*) (see Table E1 for construct items; see Table E2 for results of randomization checks). Finally, the experiment ended with a manipulation check that gauged whether the participants were aware of the availability of dockless bike-sharing services. Eight participants (3.2%) failed to pass the manipulation check, yet the results remained consistent no matter whether these participants were included or excluded in the analyses. We reported the results based on the sample of all participants (See E-Companion E4, E5, F4, and F5 for the results based on the sample excluding participants failed to pass manipulation checks).

5.2 Results and Mechanisms

We first ran a *t*-test analysis on travel time to show that it generally took more time for patients to go to hospitals than clinics ($M_{\text{Hospital}} = 29.24$, $M_{\text{Clinic}} = 11.48$, $t = 15.10$, $p < .001$). Then, a chi-square test with *Hosp_choice* as the dependent variable showed that the participants in the bike condition were more likely to choose the hospital compared to those in the non-bike condition [$M_{\text{Non-bike}} = 0.248$, $M_{\text{Bike}} = 0.408$, $\chi^2(1) = 7.259$, $p = .007$]. Specifically, 40.8% of the participants in the bike condition chose to go to the hospital, while only 24.8% of the participants did so in the non-bike condition, supporting H1.

According to our theoretical framework, perceived geographical accessibility to medical institutions and the weight

of quality in quality-accessibility trade-off served as serial mediators in the effect of dockless bike sharing on patients' choice of medical institutions (see Figure 1). To test such mediating effect, we first conducted two ANOVA tests with *Bike_sharing* as the independent variable, and *Hosp_access* and *Cli_access* as the dependent variables. The results indicated that dockless bike sharing increased the participants' perceived geographical accessibility to the hospital ($M_{\text{Non-bike}} = 4.104$, $M_{\text{Bike}} = 4.936$, $F = 13.31$, $p < .001$). However, it did not exert a significant influence on the participants' perceived geographical accessibility to the clinic ($M_{\text{Non-bike}} = 6.082$, $M_{\text{Bike}} = 6$, $F = 0.43$, $p = .511$). This finding aligned with our initial assumption that dockless bike sharing had limited impact on enhancing geographical accessibility to clinics. We then performed a similar ANOVA test using *Trade-off* as the dependent variable. The result revealed a positive and significant effect of dockless bike sharing on the weight of quality in quality-accessibility trade-off ($M_{\text{Non-bike}} = 4.571$, $M_{\text{Bike}} = 5.136$, $F = 5.94$, $p = .016$).

Next, we estimated a regression model with *Bike_sharing* as the independent variable, *Hosp_access*, *Cli_access*, and *Trade-off* as the mediators, *Hosp_choice* as the dependent variable, and participants' age, gender, education level, income level, medical literacy, perceived severity of the symptoms, and frequency of dockless bike sharing usage as the control variables (Model 6, with 5000 bootstraps, see Figure 1) (Hayes 2017). The coefficients were estimated by Partial Least Squares Structural Equation Modeling (PLS-SEM) using SmartPLS. We found a significant indirect effect of dockless bike sharing on the choice of medical institution through the path of perceived geographical accessibility to the hospital and the weight of quality in quality-accessibility trade-off ($\text{Bike_sharing} \rightarrow \text{Hosp_access} \rightarrow \text{Trade-off} \rightarrow \text{Hosp_choice}$, $\beta'_{\text{indirect}} = .058$, $p = .001$). In other words, the availability of dockless bike sharing increased patients' perceived geographical accessibility to the hospital which, in turn, augmented the weight of quality in quality-accessibility trade-off that ultimately increased the likelihood that patients chose to go to a hospital. However, no significant indirect effect was observed through the path of perceived geographical accessibility to the clinic and the weight of quality in quality-accessibility trade-off ($\text{Bike_sharing} \rightarrow \text{Cli_access} \rightarrow \text{Trade-off} \rightarrow \text{Hosp_choice}$, $\beta'_{\text{indirect}} = -.001$, $p = .910$). The direct effect of dockless bike sharing on the choice of medical institution was non-significant with the inclusion of the serial mediators ($\beta = .081$, $p = .097$, see in Figure 1), suggesting a full mediation effect. These findings enrich the literature on dockless bike sharing (e.g., Chu et al., 2021; Jin et al., 2024) by revealing the underlying mechanisms in the relationship between dockless bike sharing and disease-institution mismatch. We show that improved geographical accessibility to hospitals, due to the availability of dockless bike sharing, causes patients to rethink their quality-accessibility trade-off, and serves as the underlying

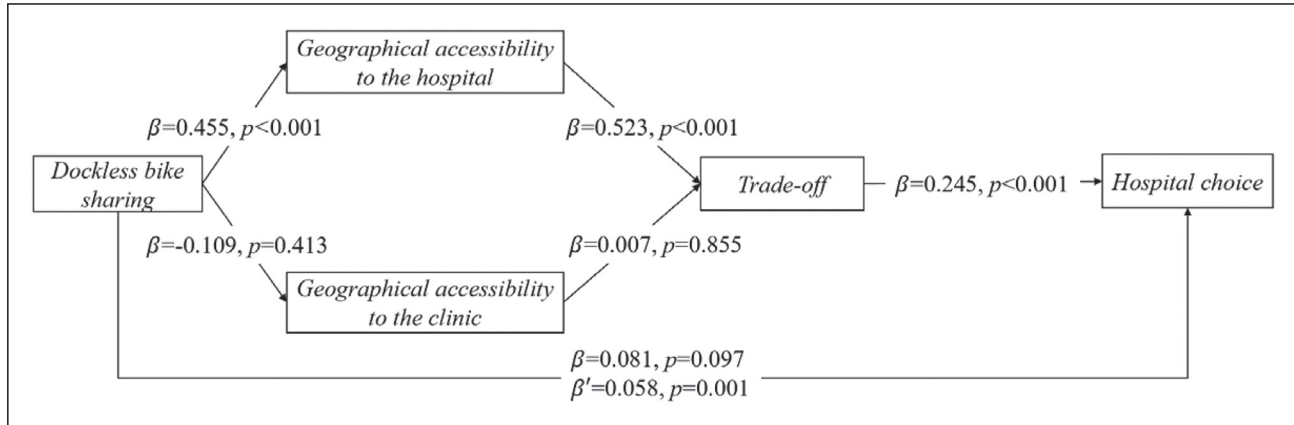


Figure 1. Empirical results: effect of dockless bike sharing on choice of medical institutions.

mechanism exacerbating the mismatch between the severity of patients' diseases and the level of medical institutions visited.

6 Study 2b

In Study 2a, participants were informed of the available transportation methods by textual information. To make the scenario more realistic, we conducted Study 2b in which participants were provided a picture (i.e., street view) and were asked to explore possible transportation methods that could be used to go to medical institutions in the picture. We leveraged Midjourney to create two street-view pictures. These two pictures were identical except that dockless shared bikes were present in the right corner of one of the pictures. In the *dockless bike sharing condition*, the picture with dockless shared bikes (see Figure 2(a)) was shown to the participants, whereas in the *non-dockless bike sharing condition*, the picture without dockless shared bikes (see Figure 2(b)) was shown. To ensure the participants had been aware of all transportation options in the picture, we asked them to list all available transportation options after viewing the picture. To further validate the assumption that people were generally farther from hospitals than clinics, we collected the information of distances from participants' home to the clinic and the hospital that they usually visited. The other procedure and measurements used in Study 2b were the same as those in Study 2a (see Table E2 for the results of randomization checks). Two hundred and fifty subjects (68.8% female, $M_{\text{age}} = 30.70$) recruited from Credamo participated in Study 2b. Eleven participants (4.4%) failed to pass the manipulation check. The results remained consistent no matter whether these participants were included or excluded in the analyses. We reported the results based on the sample of all participants.

Overall, the results of Study 2b were similar to those of Study 2a. First, the t -test result showed that it usually took more time for patients to go to hospitals than clinics ($M_{\text{Hospital}} = 19.92$, $M_{\text{Clinic}} = 8.84$, $t = 16.30$, $p < .001$). A

chi-square test with *Hosp_choice* as the dependent variable showed that the participants in the bike condition were more likely to choose the hospital compared to those in the non-bike condition [$M_{\text{Non-bike}} = 0.464$, $M_{\text{Bike}} = 0.616$, $\chi^2(1) = 5.81$, $p = .016$], supporting H1.

Regarding the mediating effects, the ANOVA tests revealed that dockless bike sharing increased the participants' perceived geographical accessibility to the hospital ($M_{\text{Non-bike}} = 4.757$, $M_{\text{Bike}} = 5.349$, $F = 10.74$, $p = .001$), but it did not exert a significant influence on the participants' perceived geographical accessibility to the clinic ($M_{\text{Non-bike}} = 6.016$, $M_{\text{Bike}} = 6.168$, $F = 2.58$, $p = .110$). Additionally, we found a positive and significant effect of dockless bike sharing on the weight of quality in quality-accessibility trade-off ($M_{\text{Non-bike}} = 5.163$, $M_{\text{Bike}} = 5.563$, $F = 4.23$, $p = .041$). After adding two control variables (i.e., distance to the hospital and distance to the clinic) to the regression model used in Study 2a, we found a significant indirect effect of dockless bike sharing on the choice of medical institution through the path of perceived geographical accessibility to the hospital and the weight of quality in quality-accessibility trade-off ($\text{Bike_sharing} \rightarrow \text{Hosp_access} \rightarrow \text{Trade-off} \rightarrow \text{Hosp_choice}$, $\beta'_{\text{indirect}} = .044$, $p = .008$). However, no significant indirect effect was observed through the path of perceived geographical accessibility to the clinic and the weight of quality in quality-accessibility trade-off ($\text{Bike_sharing} \rightarrow \text{Cli_access} \rightarrow \text{Trade-off} \rightarrow \text{Hosp_choice}$, $\beta'_{\text{indirect}} = -.002$, $p = .635$). The direct effect of dockless bike sharing on the choice of medical institution was non-significant with the inclusion of the serial mediators ($\beta = .079$, $p = .127$, see in Figure 3), suggesting a full mediation effect.

7 Study 3a

The first goal of Study 3a was to replicate the findings regarding the effects of dockless bike sharing on the number of medical tests in hospitals and clinics under a randomized controlled experimental setting. Similar to Study 2a, we manipulated the availability of dockless bike sharing by designing

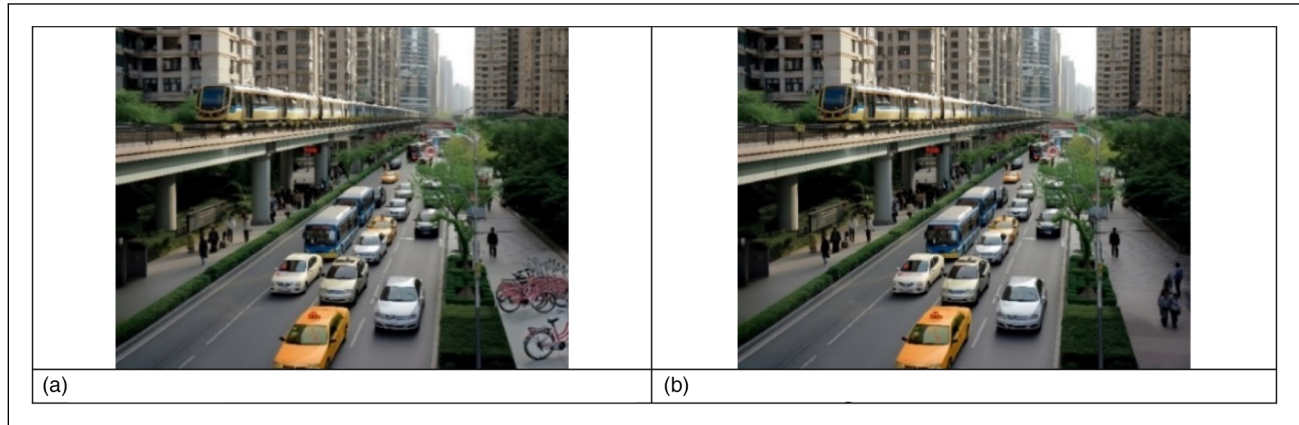


Figure 2. Street view pictures for dockless bike sharing and non-dockless bike sharing conditions. (a) A street view picture with dockless shared bikes; (b) A street view picture without dockless shared bikes.

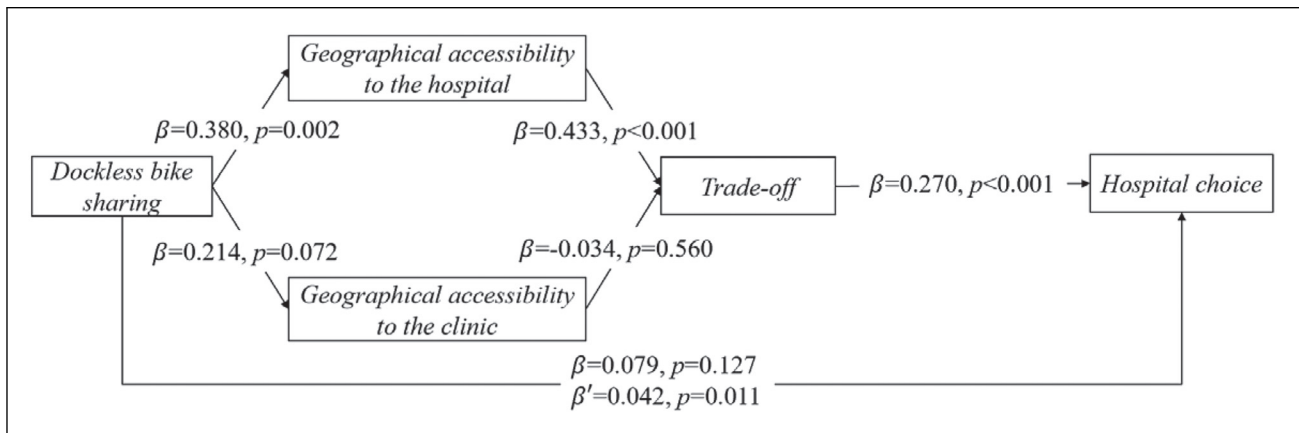


Figure 3. Empirical results: effect of dockless bike sharing on choice of medical institutions.

a dockless bike sharing condition and a non-dockless bike sharing condition, which enabled us to infer a causal relationship between dockless bike sharing and the number of medical tests per visit. The second goal was to examine the underlying mechanism behind such relationship. In accordance with our theorizing, the availability of dockless bike sharing was expected to improve geographical accessibility to medical institutions which, in turn, lowered patients' perceived necessity of medical tests, thereby leading to a decrease in the number of medical tests per visit.

7.1 Experiment Design and Procedure

Five hundred and twenty subjects (64.42% female, $M_{\text{age}} = 30.823$) recruited from Credamo participated in Study 3a in exchange for monetary compensation. The study consisted of a 2 (medical institution: hospital vs. clinic) $\times$ 2 (transportation: dockless bike sharing vs. non-dockless bike sharing) between-subjects design. Participants were randomly assigned to one of the four experiment conditions.

As in Study 2a, participants first answered questions about their demographics (i.e., gender, age, education level, and income level). Subsequently, participants were asked to imagine a scenario in which they had several symptoms (i.e., a stuffy nose, runny nose, and cough).⁹ They were told that they went to a hospital that they usually visit in the *hospital condition*, or a clinic that they usually visit in the *clinic condition*. In all conditions, participants were informed that, based on their symptoms, a physician had recommended some basic medical tests for them. Besides the basic medical tests, they might request additional medical tests, such as a chest computed tomography (CT) scan (a diagnostic imaging test providing detailed images of internal organs, bones, soft tissue, and blood vessels), to rule out potential complications. While a chest CT scan was not deemed necessary at the moment, if their symptoms persisted for several days, they might need to revisit the medical institution for the test. In the *dockless bike sharing condition*, participants were informed that they could use any transportation methods (including bus, subway, taxi, bike, dockless shared bike, motorbike, and

private car) or walk to go to the medical institution. By contrast, in the *non-dockless bike-sharing condition*, they could use any transportation method (including bus, subway, taxi, bike, motorbike, and private car) or walk to go to the medical institution.

Participants first reported whether they would like to request any additional medical tests during this visit (*Test_request*). They then rated their perceived necessity of additional medical tests (*Test_necessity*) and perceived geographical accessibility to the medical institution (*Access*). They also answered questions about their medical literacy (*Med_literacy*), perceived severity of the symptoms in the given scenario (*Severity*), travel time to the medical institution (*Trv_time*), and frequency of dockless bike sharing usage (*Bike_freq*) (see Table F1 for construct items; see Table F2 for results of randomization checks). Finally, the experiment ended with a manipulation check that gauged whether the participants were aware of the availability of dockless bike sharing. Ten participants (1.9%) failed to pass the manipulation check, yet the results remained consistent with or without these participants. We reported the results based on the sample of all participants.

7.2 Results and Mechanisms

We first ran a *t*-test analysis on travel time to show that it usually took more time for patients to go to hospitals than clinics ($M_{\text{Hospital}} = 28.70$, $M_{\text{Clinic}} = 19.98$, $t = 8.42$, $p < .001$). We then conducted an analysis of covariance (ANCOVA) test with *Bike_sharing* as the independent variables, *Test_request* as the dependent variable, and participants' age, gender, education level, income level, medical literacy, perceived severity of the symptoms, medical institution type, and frequency of dockless bike sharing usage as the covariates. The results showed that the overall effect of dockless bike sharing on requesting additional medical tests was significant ($M_{\text{Non-bike}} = 0.673$, $M_{\text{Bike}} = 0.596$, $F = 5.36$, $p = .021$).

Given that dockless bike sharing improved patients' accessibility to hospitals and clinics to different extents, we performed similar ANCOVA tests based on the hospital and the clinic subsamples, respectively. Consistent with the findings of Study 1, the results revealed that participants in the *hospital plus bike condition* were less likely to request additional medical tests, compared to those in the *hospital plus non-bike condition* ($M_{\text{Non-bike}} = 0.746$, $M_{\text{Bike}} = 0.569$, $F = 10.55$, $p = .001$). However, the availability of dockless bike sharing had no significant effect on requesting additional medical tests in clinics ($M_{\text{Non-bike}} = 0.6$, $M_{\text{Bike}} = 0.623$, $F = 0.07$, $p = .786$). Therefore, H2 was partially supported.

Since dockless bike sharing did not exert an effect on requesting additional medical tests in clinics, we explored the mechanism underlying the negative effect of dockless bike sharing on requesting additional medical tests in hospitals. According to our theorizing, perceived geographical accessibility to the hospital and perceived necessity of

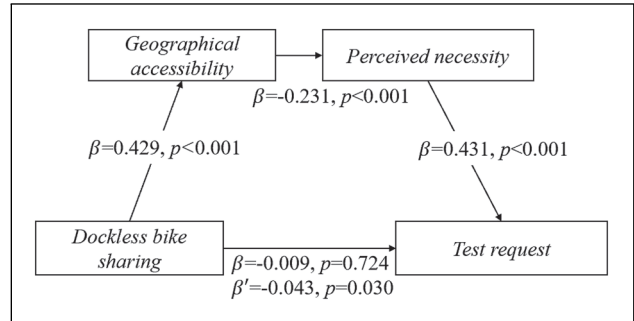


Figure 4. Empirical results: effect of dockless bike sharing on requesting additional medical tests in hospitals.

additional medical tests served as serial mediators in the effect of dockless bike sharing on the number of medical tests per hospital visit (see Figure 4). To test the mediating effect, we first conducted an ANOVA test with *Bike_sharing* as the independent variable and *Access* as the dependent variable. The result revealed that participants in the *bike condition* reported a significantly higher perceived geographical accessibility to the hospital than those in the *non-bike condition* ($M_{\text{Non-bike}} = 4.828$, $M_{\text{Bike}} = 5.503$; $F = 12.56$, $p < .001$). We then conducted a similar ANOVA test with *Test_necessity* as the dependent variable. The results showed a negative and significant effect of dockless bike sharing on participants' perceived necessity of additional medical tests ($M_{\text{Non-bike}} = 4.996$, $M_{\text{Bike}} = 4.265$; $F = 9.82$, $p = .002$).

Next, we estimated a regression model with *Bike_sharing* as the independent variable, *Access* and *Test_necessity* as the serial mediators, *Test_request* as the dependent variable, and participants' age, gender, education level, income level, medical literacy, perceived severity of the symptoms, and frequency of dockless bike sharing usage as the control variables (Model 6, with 5000 bootstraps, see Figure 4) (Hayes 2017). The coefficients were estimated by PLS-SEM using SmartPLS. The results revealed a significant indirect effect of dockless bike sharing on requesting additional medical tests through the path of increased perceived geographical accessibility to the hospital and decreased perceived necessity of additional medical tests ($\text{Bike_sharing} \rightarrow \text{Access} \rightarrow \text{Test_necessity} \rightarrow \text{Test_request}$, $\beta'_{\text{indirect}} = -.043$, $p = .030$). Furthermore, the direct effect of dockless bike sharing on requesting additional medical tests was non-significant with the inclusion of the serial mediators (*Test_request*: $\beta = -.009$, $p = .724$, see Figure 4), suggesting a full mediation effect.

While scholars have examined how to reduce the overuse of medical tests through lowering the cost of information sharing or regulating physicians' work environment (e.g., Ayabakan et al., 2017; Young et al., 2023), our findings reveal the underlying mechanism in how dockless bike sharing reduces the overuse of medical tests. Specifically, the availability of dockless bike sharing improved patients' perceived geographical accessibility to hospitals which, in turn, lowered their perceived necessity of additional medical tests, thereby leading

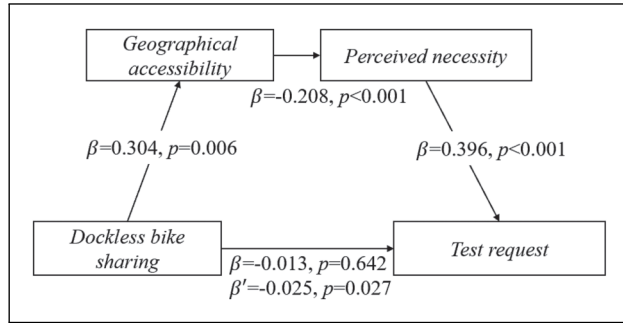


Figure 5. Empirical results: effect of dockless bike sharing on requesting additional medical tests in hospitals.

to a reduction in the number of unnecessary medical tests in hospitals. Based on this mechanism, the government may also promote other shared mobility services, such as ride sharing, so that unnecessary medical tests would be reduced with enhanced geographical accessibility to hospitals.

8 Study 3b

Similar to Study 2b, AI-generated street view pictures (see Figure 5) instead of textual information were used in Study 3b to make the scenario more realistic. Moreover, the additional medical test example, a chest CT scan, was replaced with a blood test in Study 3b, because participants with low medical literacy might not be able to fully understand what a CT scan was.¹⁰ We also collected the information of distances from participants' home to the medical institution that they usually visited. Except for these differences, the other procedures and measurements were the same as those in Study 3a (see Table F2 for the results of randomization checks). Five hundred and twenty subjects (66.9% female, $M_{\text{age}} = 30.710$) recruited from Credamo participated in Study 3b. Sixteen participants (3.1%) failed to pass the manipulation check, yet the results remained consistent with or without these participants. We reported the results based on the sample of all participants.

As in Study 3a, the t -test result revealed that it usually took more time for patients to go to hospitals than clinics ($M_{\text{Hospital}} = 19.87$, $M_{\text{Clinic}} = 14.78$, $t = 5.46$, $p < .001$). The ANCOVA test showed that the overall effect of dockless bike sharing on requesting additional medical tests was significant ($M_{\text{Non-bike}} = 0.769$, $M_{\text{Bike}} = 0.677$, $F = 3.93$, $p = .048$). Similar to Study 3a, the subsample analysis results suggested that participants in the *hospital plus bike condition* were less likely to request additional medical tests, compared to those in the *hospital plus non-bike condition* ($M_{\text{Non-bike}} = 0.777$, $M_{\text{Bike}} = 0.638$, $F = 4.46$, $p = .036$). Nevertheless, the availability of dockless bike sharing had no significant effect on requesting additional medical tests in clinics ($M_{\text{Non-bike}} = 0.762$, $M_{\text{Bike}} = 0.715$, $F = 0.27$, $p = .604$).

Regarding the underlying mechanism, we first conducted ANOVA tests to show a positive and significant effect of

dockless bike sharing on participants' perceived geographical accessibility ($M_{\text{Non-bike}} = 5.590$, $M_{\text{Bike}} = 5.944$; $F = 7.97$, $p = .005$) and a negative and significant effect of dockless bike sharing on participants' perceived necessity of additional medical tests ($M_{\text{Non-bike}} = 5.058$, $M_{\text{Bike}} = 4.542$; $F = 5.97$, $p = .015$). Next, after adding distance to the hospital as a control variable to the regression model used in Study 3a, we found a significant indirect effect of dockless bike sharing on requesting additional medical tests through the path of increased perceived geographical accessibility to the hospital and decreased perceived necessity of additional medical tests ($\text{Bike_sharing} \rightarrow \text{Access} \rightarrow \text{Test_necessity} \rightarrow \text{Test_request}$, $\beta'_{\text{indirect}} = -.025$, $p = .027$). Furthermore, the direct effect of dockless bike sharing on requesting additional medical tests was non-significant with the inclusion of the serial mediators (Test_request : $\beta = -.013$, $p = .642$, see Figure 5), suggesting a full mediation effect.

9 General Discussion

9.1 Theoretical Contributions

This study has several important theoretical contributions. First, this is an initial effort to explore the ineffective use of medical resources from the perspective of patient decision-making. While previous OM literature has focused on mitigating ineffective medical resource usage, such as inappropriate diagnostic imaging (Young et al., 2023), delay in patient transfer (Lu and Lu, 2018), and misplacement of patients (Bretthauer et al., 2011; Liu et al., 2019), from the perspective of healthcare providers, this study goes beyond past research by highlighting patients' choices in medical resource management. Importantly, we demonstrate that patients' healthcare decision-making can be greatly influenced by their geographical accessibility to medical institutions. Although accessibility to medical institutions is a critical aspect of medical service operations (de Vries et al., 2020; Tipping et al., 2022), the last-mile problem in accessing medical institutions has received little research attention. Our research indicates that, like other public transportation users, patients with LRD may also encounter the "last-mile problem" when they rely on public transportation to reach medical institutions. Thus, our work contributes to the OM literature by linking the last-mile problem to patients' medical choices and uncovering the important role of geographical accessibility in their medical decision-making.

Second, our research represents a preliminary effort to establish theoretical connections between dockless bike sharing and two key issues in medical resource management, namely disease-institution mismatch and the overuse of medical tests. Although the OM literature has investigated dockless bike-sharing operations (Freund et al., 2018; He et al., 2021; Qi and Shen, 2019) and medical resource management (Ayer et al., 2019; Chan et al., 2019), few studies have bridged these two areas to explore the impact of dockless bike sharing on the ineffective use of medical resources. Our research

advances the sharing economy and medical resource management literature by demonstrating that dockless bike sharing can exacerbate disease-institution mismatch while alleviating the overuse of medical tests. We show that, beyond internal factors, external factors, such as transportation options, can also play a significant role in shaping patients' medical choices.

Third, this study enriches the OM literature by revealing the psychological mechanisms underlying the effects of dockless bike sharing on two types of ineffective medical resource use. Different from outcomes like housing prices or public transportation usage (Chu et al., 2021; Jin et al., 2024), patients' medical decision-making is shaped by more complex psychological dynamics—an area that has received less attention in the OM literature. By introducing the concept of quality-accessibility trade-off in selecting medical institutions, we demonstrate that enhanced geographical accessibility brought by dockless bike sharing disrupts the balance between quality and accessibility, prompting patients with LRD to select higher-ranked hospitals. Moreover, by introducing the concept of perceived necessity in requesting medical tests, we show that dockless bike sharing increases geographical accessibility to hospitals which, in turn, attenuates the perceived necessity of medical tests, and ultimately mitigates the number of unnecessary medical tests. Our novel psychological understanding of the changes in geographical accessibility advances the OM literature by highlighting the importance of considering patients' expectation of geographical accessibility in both bike-sharing and medical service operations.

9.2 Managerial Implications

This research offers several important managerial implications for governments, shared mobility platforms, and healthcare providers. While previous studies have demonstrated the positive influences of dockless bike sharing on complementing public transportation (Jin et al., 2024) and reducing housing price premium (Chu et al., 2021), this study reminds us that dockless bike sharing can also lead to less optimal use of medical resources, such as exacerbating the mismatch between the severity of a patient's condition and the ranking of medical institution they visit. Considering the mediating role of geographical accessibility and patients' quality-accessibility trade-off in this relationship, a feasible approach is to impose interventions on the mediating pathway to prevent dockless bike sharing from exacerbating the disease-institution mismatch. In other words, although dockless bike sharing improves geographical accessibility to hospitals, if the enhanced accessibility does not lead to more emphasis on quality in the quality-accessibility trade-off, the issue of disease-institution mismatch would not be aggravated by dockless bike sharing. Specifically, governments can address this issue by launching educational programs to disseminate basic medical knowledge. When making quality-accessibility trade-offs, patients with LRD prefer hospitals over clinics

due to limited understanding of their health conditions and concerns about the adequacy of treatment at clinics. Enhancing awareness of common health issues can help individuals feel less anxious about mild symptoms. As a result, patients with minor conditions would be less inclined to prioritize the ranking of medical institutions and would continue visiting clinics, even if higher ranked hospitals become more geographically accessible after the introduction of dockless bike sharing.

Another possible approach to mitigating the disease-institution mismatch is to charge additional fees for parking shared bikes near hospital entrances. Building on the concept of geo-fencing proposed by He et al. (2021), dockless bike-sharing platforms can designate no-parking zones near prestigious hospitals on the map. Such a strategy has been implemented in some large cities in China (see Figure G2 in E-companion G for an example of no-parking zones near a prestigious hospital). Users have to pay an additional dispatch fee of approximately 0.7 USD if they leave their shared bikes in a no-parking zone. Since only patients with LRD may go to hospitals by bike, raising the cost of using dockless shared bikes for hospital visits can encourage these patients to choose clinics instead. However, the relatively low dispatch fee still allows them to access high-quality medical resources at a slightly higher cost. Instead of increasing the cost of accessing high-quality medical resources, an alternative approach is to offer coupons or discounts to bike-sharing users who select clinics as their destinations.

In addition, the findings regarding the overuse of medical tests indicate that patients with LRD make more rational requests for medical tests after the availability of dockless bike sharing, as it increases their anticipated geographical accessibility to medical institutions for future visits. Overall, dockless bike sharing acts as a “double-edged sword” in terms of its impact on the ineffective use of medical resources—simultaneously exacerbating disease-institution mismatch and alleviating the overuse of medical tests. To reconcile the conflicting managerial implications derived from the two sides of dockless bike sharing, we provide suggestions grounded in the underlying mechanism of improved geographical accessibility.

As per our theorizing, geographical accessibility to medical institutions can influence medical resource usage for all patients, including those with LRD and HRD. While dockless bike sharing primarily benefits patients with LRD, ride-sharing services are more commonly used by patients with HRD to seek treatment in prestigious hospitals. Thus, the government can advocate for ride-sharing services to improve geographical accessibility to prestigious hospitals for patients with HRD, ensuring an optimal match between the severity of patients' conditions and the level of medical institutions they visit. This approach can also help reduce the number of unnecessary medical tests requested by patients with HRD. Moreover, ride-sharing platforms can increase vehicle availability near

hospitals or collaborate with healthcare facilities to offer specialized transportation services tailored to the needs of patients with HRD.

Another possible resolution for alleviating the overuse of medical tests involves facilitating healthcare coordination across hospitals and clinics through health information sharing (Atasoy et al., 2021) and granting patients who initially undergo treatment at hospitals the choice of receiving medical tests later at a nearby clinic. In this case, patients would not request excessive medical tests in the current visit as they anticipate a heightened level of geographical accessibility to medical institutions for future visits.

9.3 Limitations and Future Research

Our findings need to be interpreted in the context of the limitations of this study, which also provide opportunities for further research. First, due to the data limitation, we only tested the hypotheses using patients' EHR in an urban area of China. However, patients in the rural area may not have the last-mile problem, and thus dockless shared bikes may not play an important role in changing these patients' medical resource usage. To increase the generalizability of our findings, future research can verify the hypotheses using data from other areas. Second, this research focuses on the impact of dockless bike sharing on two types of ineffective use of medical resources. Other medical-related decisions, such as selection of doctors and revisit regularity, may also be affected by dockless bike sharing. Future research can examine how dockless bike sharing influences these outcomes. Third, since both dockless bike sharing and ride sharing can enhance geographical accessibility, future research can explore the strategies to optimize the allocation of shared bikes and vehicles near hospitals and clinics to effectively reduce the overuse of medical tests without aggravating the disease-institution mismatch.

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Supplemental Material

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Notes

1. <https://www.crt.org.cn/details.html?id=11&contentId=2922> (Access on 30 December 2024).
2. <https://www.thatsmags.com/beijing/post/20767/massive-piles-of-shared-bikes-block-hospital-entrances-in-beijing> (Access on 30 December 2024).
3. This effect mainly applies to patients with LRD because patients with HRD have a higher possibility of motion disorder and are unlikely to be capable of riding a bike to a hospital (Dorsey et al., 2013). For this reason, the entry of dockless bike sharing is unlikely to influence patients with HRD's choice of medical institutions.
4. We also cross-validated the disease category based on Alipay Severe Disease Insurance.
5. The treatment group's upper age limit was set at 25 to ensure a balanced age range (i.e., eight years) between the treatment and control groups.
6. According to the China Health Statistical Yearbook 2022, the morbidity rates in eastern China for various age groups are as follows: 18.5% (aged 0–4), 11.4% (aged 5–14), and 10.7% (aged 15–24) in urban areas, and 22.3% (aged 0–4), 12.6% (aged 5–14), and 10.8% (aged 15–24) in rural areas.
7. We conducted a robustness check by incorporating the burn-in period into the post-treatment period (pre-treatment period: March 2017 to August 2017; post-treatment period: September 2017 to September 2018), and found similar results (see Model (C23) and Model (C24) of Table C3 in E-companion C3).
8. Participants were informed that they already had a polymerase chain reaction test to eliminate the possibility of COVID-19.
9. Participants were informed that they already had a polymerase chain reaction test to eliminate the possibility of COVID-19.
10. We thank an anonymous reviewer for proposing this concern.

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